

1

The table summarizes the UV index value recorded by a research assistant at noon each day for 48 days.

UV index	Number of days
1	8
2	13
3	12
4	15

According to the table, a UV index value of 1 was recorded on how many days?

☐ (A) 48

☐ (B) 40

☐ (C) 8

☐ (D) 4

3

The speed of a white-throated needletail, a type of bird, in flight was measured to be 79 miles per hour. What was the white-throated needletail's measured speed, in kilometers per hour? (Use 1 mile = 1.6 kilometers.)

☐ (A) 126.4

☐ (B) 80.6

☐ (C) 77.4

☐ (D) 49.4

2

The function j is defined by $j(x) = \frac{x}{7}$. What is the value of $j(42)$?

☐ (A) 42

☐ (B) 35

☐ (C) 7

☐ (D) 6

4

$$(2x^2 + 8x - 6) - (4x^2 + 7x)$$

The given expression is equivalent to $-2x^2 + ax - 6$, where a is a constant. What is the value of a ?

5

What is the y -intercept of the graph of $y = 11^x$ in the xy -plane?

☐ (A) (0, 1)

☐ (B) (0, 0)

☐ (C) (0, 11)

☐ (D) (0, 10)

7

To cut a lawn, Antwan charges a fee of \$15.00 for his equipment and \$8.50 per hour spent cutting a lawn. Taylor charges a fee of \$12.00 for her equipment and \$9.00 per hour spent cutting a lawn. If x represents the number of hours spent cutting a lawn, what are all the values of x for which Taylor's total charge is greater than Antwan's total charge?

☐ (A) $5 \leq x \leq 6$
☐ (B) $6 \leq x \leq 7$
☐ (C) $x < 5$
☐ (D) $x > 6$

6

$$y = x + 19$$

$$y = -x + 33$$

The solution to the given system of equations is (x, y) . What is the value of $2y$?

☐ (A) 14

☐ (B) 26

☐ (C) 52

☐ (D) 104

8

The table summarizes the number of objects in each group.

Group	Number of objects
A	375
B	54
C	690
D	81
Total	1,200

The number of objects in group C is $p\%$ of the number of objects in group A. What is the value of p ?

9

Which quadratic equation has exactly one distinct real solution?

(A) $-97x^2 = 0$

(B) $9x^2 + 3 = 0$

(C) $9x^2 + 3x + 1 = 0$

(D) $-97x^2 - 97x = 0$

10

$$x^2 + 9x + 5 = 0$$

One solution to the given equation can be written as

$$x = \frac{-9 + \sqrt{k}}{2}, \text{ where } k \text{ is a constant. What is the value}$$

of k ?

11

The table gives the distribution of favorite core subject and grade level for 80 students.

Subject	Grade level			
	Sixth	Seventh	Eighth	Total
English	4	9	9	22
Mathematics	10	1	9	20
Science	4	6	4	14
Social studies	6	9	9	24
Total	24	25	31	80

If a student is selected at random, what is the probability of selecting a student whose favorite core subject is mathematics?

(A) $\frac{1}{10}$

(B) $\frac{1}{5}$

(C) $\frac{1}{4}$

(D) $\frac{2}{3}$

12

If $17 - 5(4 - 7x) = 4 - 6(4 - 7x)$, what is the value of $4 - 7x$?

13

A rectangular poster has an area of 360 square inches. A copy of the poster is made in which the length and width of the original poster are each increased by 40%. What is the area of the copy, in square inches?

14

The function f is defined by $f(x) = 41(0.28)^x$. For any positive integer n , the value of $f(n)$ is $p\%$ less than the value of $f(n - 1)$. What is the value of p ?

(A) 72

(B) 59

(C) 41

(D) 28

15

$$5x = 75y - 125$$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

(A) $x = 3y$

(B) $\frac{1}{5}x = 3y$

(C) $x = 15y - 25$

(D) $\frac{1}{5}x = 15y - 25$

16

$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(35) > 0$, $f(38) < 0$, and $f(41) > 0$, which of the following could be the value of $a + b$?

(A) 38

(B) 40

(C) 73

(D) 75

17

$$\sqrt{x^2 - 20x + 100} - 11x = 0$$

What are all possible solutions to the given equation?

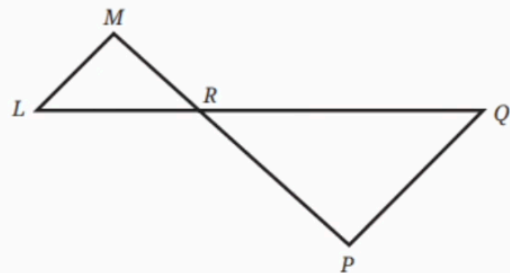
(A) -1

(B) $\frac{5}{6}$

(C) -1 and 1

(D) -1 and $\frac{5}{6}$

18



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$ and $\overline{RP}$ are 9 and 17 units, respectively. The area of $\triangle LMR$ is 45 square units. What is the area of $\triangle PQR$, in square units?

(A) $\frac{3.645}{289}$

(B) $\frac{405}{17}$

(C) 85

(D) $\frac{1,445}{9}$

19

$$6x^2(3x - 16)(3x - u) = 0$$

In the given equation, u is a positive constant. The sum of the solution to the equation is 13. What is the value of u ?

20

The cost of renting a power washer is \$44 for the first day and \$22 for each additional day. Which of the following functions gives the cost $C(d)$, in dollars, of renting the power washer for d days, where d is a positive integer?

(A) $C(d) = 22d + 22$

(B) $C(d) = 22d + 44$

(C) $C(d) = 44d - 22$

(D) $C(d) = 44d + 66$

21

Function f is a quadratic function. The graph of $y = f(x)$ in the xy -plane has a vertex at $(9, -7)$, contains the point $(8, -9)$, and has a y -intercept at $(0, a)$. The graph of $y = 6 \cdot f(x)$ has a y -intercept at $(0, b)$. What is the positive difference between a and b ?

22

A 9-pound cat eats two types of canned cat food: chicken-flavored and duck-flavored. The recommended amount of chicken-flavored food is 2.25 cans per 8 pounds of a cat's weight per day. The recommended amount of duck-flavored food is 0.75 cans per 5 pounds of a cat's weight per day. If c is the number of chicken-flavored cans and d is the number of duck-flavored cans a 9-pound cat eats in one day, which equation describes all the possible values of c and d for the cat to eat the recommended amount of food based on its weight?

(A) $\frac{0.75}{5}c + \frac{2.25}{8}d = 9$

(B) $\frac{2.25}{8}c + \frac{0.75}{5}d = 9$

(C) $\frac{5}{0.75}c + \frac{8}{2.25}d = 9$

(D) $\frac{8}{2.25}c + \frac{5}{0.75}d = 9$